

SURAJ SURVE

GATE DA Qualified | AI & Robotics Engineer | Full-Stack · ML · Cloud · DevOps

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EDUCATION

B.Tech – Artificial Intelligence & Data Science (Pursuing)

2023 – 2027

Marathwada Mitra Mandal's College of Engineering, Pune | Grade: A

Activities & Societies: NEURA MMCOE (Vice President) · Robocon Team Rudra (AI Engineer) · Cycling

CBSE HSC – Dhruv Global School, Sangamner, Ahilyanagar

2023 | 80%

CBSE SSC – Vasundhara Academy, Akole, Ahilyanagar

2021 | 95%

TECHNICAL SKILLS

Robotics & Embedded: ROS2, MoveIt, Gazebo, Digital Twin, 6-DOF Arm, Bus Servo, Raspberry Pi, Arduino, ESP32, SLAM, Path Planning, Microsoft AirSim

Deep Learning & CV: PyTorch, TensorFlow/Keras, YOLO (v8/v11/v12/World/Nano), OpenCV, LSTM, Attention Mechanisms, Sensor Fusion (LiDAR/Camera), ONNX, Model Quantization

Reinforcement Learning: Double DQN (N-step), Hindsight Experience Replay, Curriculum Learning, Curiosity-Driven Exploration, CUDA, TensorBoard

Generative AI & LLMs: RAG Pipelines, Google Gemini 2.5 Flash, Ollama, DeepSeek Coder, OpenAI GPT, Fine Tuning, Prompt Engineering (CoT), NL2SQL/NL2Python, ChromaDB, Pinecone, Vector Embeddings

Mobile & Full Stack: Flutter/Dart, React 19, Next.js 16, TypeScript, Node.js, Express.js, FastAPI, Flask, Tailwind CSS 4, REST APIs, Microservices, Swagger/OpenAPI

Databases: PostgreSQL / Supabase (RLS, Edge Functions, Realtime), MongoDB, MySQL, Firebase Auth, ChromaDB

Cloud, DevOps & MLOps: AWS (IoT Core), GCP, Docker & Compose, GitLab CI/CD, Fastlane (Play Store), Render, Vercel, Prefect, Git, Linux, Edge AI (Raspberry Pi)

Data Science: Time-Series Forecasting, OLS Regression, IQR, Feature Engineering, Pandas, NumPy, SciPy, Matplotlib, Seaborn

Languages: Python (Advanced), Dart, TypeScript, C, C++, SQL, JavaScript, Bash/Shell, PowerShell

EXPERIENCE

AI Robotics Engineer · M CAD Solutions (Mentor: Manish Puddar) · Internship

Feb 2026 – Present

- Designed & deployed a 6-DOF robotic arm (3D-printed, bus servo motors) with full stack: ROS2, MoveIt, Gazebo, C++ & Python; created Digital Twin with real-time bidirectional sync for safe task pre-validation.
- Built Brain Node integrating LLMs (Ollama) + YOLO World: plain-English commands → object detection → MoveIt path planning → physical execution — end-to-end natural language to robot action pipeline.
- Implemented dual-camera 3D spatial mapping (YOLO Nano/Standard/World), kinesthetic learning (teach-by-demonstration waypoint replay), and predictive maintenance via continuous servo telemetry monitoring.
- Developed full operator dashboard (coordinate/keyboard/PX4 input, live telemetry) and Flutter mobile app; architecting AWS IoT Core migration for cloud-connected multi-robot fleet management.

Artificial Intelligence Engineer · Foundation for MakeltHappen Center (FMCII) · Internship

Dec 2025 – Present

- GrowBox AI gardening platform: architected FastAPI + Supabase (PostgreSQL/RLS) microservices for 498+ plant species database; engineered AI inference pipelines for real-time plant identification and disease diagnosis.
- Built proprietary recommendation algorithm scoring plant suitability by climate, experience, and agricultural parameters; deployed on Render + AWS with CI/CD workflows and Swagger UI documentation.

Artificial Intelligence Intern · Infosys Springboard · Remote

Dec 2025 – Apr 2026

- Engineered Smart Energy Forecasting system using LSTM networks achieving 95% prediction accuracy on 6 months of high-frequency IoT data; deployed production Flask web app with real-time energy-saving recommendations.
- Architected time-series data pipeline (lag variables, rolling averages, missing value imputation) ensuring high data fidelity for deep learning model training on sensor streaming data.

Artificial Intelligence Engineer · Robocon Team Rudra · On-site

Nov 2025 – Jan 2026

- Integrated computer vision (YOLO) and AI decision-making modules into competitive Robocon robotics systems for real-time autonomous object detection and task execution under competition constraints.

Vice President · [NEURA MMCOE](#) · [College AI/ML Club](#)

Jul 2024 – Jul 2025

- Led MMCOE's AI/ML student club; organized workshops, hackathons, and peer-learning sessions on deep learning, computer vision, and robotics; coordinated speaker events and student project showcases for 100+ members.

PROJECTS**GrowBox — AI Smart Gardening Platform | 382 users · 429 AI diagnoses served** [Solo Full-Stack Developer](#)

Tech: Flutter 3.x, Dart, Supabase (Auth/PostgreSQL/RLS/Edge Functions), Next.js 16, React 19, Google Gemini 2.5 Flash-Lite, PlantNet API, Paystack, AdMob, GitLab CI/CD, Fastlane

- Built production cross-platform app (v1.0.2, 39K+ LOC) with 21 screens, 16 service classes, and a 406-plant reference database; integrated Google Gemini 2.5 Flash-Lite Vision API + PlantNet for real-time AI plant disease diagnosis delivering treatment plans with confidence scoring.
- Designed 15-table PostgreSQL schema on Supabase with Row Level Security on all tables, JSONB-based AI result storage, dual-ID user pattern, and a PL/pgSQL security definer RPC for Apple Guideline 5.1.1-compliant 13-step cascading account deletion across 9 tables.
- Engineered gamification engine: watering streaks with progressive points algorithm (27 badges across 4 categories), global leaderboard, automated badge celebrations — plus OpenWeatherMap integration for location-based frost/heat/storm alerts with 3 configurable severity levels.
- Integrated Paystack live payment processing, daily 3-credit AI scan metering with composite-unique upsert idempotency, Google AdMob for freemium monetization, and timezone-aware push notifications with 4 Android channels and reminder sync engine post-restart.
- Built Next.js 16 admin dashboard (13 modules, 9 chart types, 8 live KPIs, 30-second auto-refresh) with Server Actions; configured 4-stage GitLab CI/CD pipeline (Analyze → Test → Build → Deploy) with Fastlane-automated keystore-signed APK/AAB Play Store deployment.

MessMaster Pro — Enterprise Hospitality SaaS (HospitalityOS)

[Solo Full-Stack Developer](#)

Tech: React 19, TypeScript, Vite 7, Tailwind CSS 4, Node.js, Express 4, PostgreSQL (36 tables), Supabase Realtime, JWT, Zustand, TanStack Query, PDFKit, Vercel

- Architected full-stack multi-tenant SaaS with 31 frontend pages, 21 REST API route modules (240KB+ backend logic), and a 36-table PostgreSQL schema with UUID PKs, 12 performance indexes, cascading FK, and JSONB columns — supporting both Mess Management and Restaurant modes.
- Engineered Recipe Engine auto-deducting ingredients from mess inventory on every POS order via dish_ingredients junction mapping, with full audit trail in inventory_txns and reorder-threshold alerting; dual inventory system (Central Store + per-mess local stock) with dispatch variance tracking.
- Built Restaurant Mode with table management, QR-code customer ordering (public endpoints + RLS), Kitchen Display System with real-time Supabase Realtime ticket queue, and order state machine with order_status_log audit table tracking from/to status, change source, and reasons.
- Implemented multi-tenant data isolation via JWT org_id scoping across all 21 API modules; workforce module with shift scheduling, daily attendance check-in/out, and auto-calculated payroll; auto-generated monthly invoices with attendance-based billing and multi-dimensional revenue analytics.

6-DOF AI Robotic Arm with Digital Twin

[AI Robotics Engineer @ MCAD Solutions](#)

Tech: ROS2, MoveIt, Gazebo, YOLO World/Nano, Ollama LLM, Flutter, AWS IoT Core, Raspberry Pi, C++, Python

- Brain Node: LLM (Ollama) + YOLO World converts plain-English commands → object detection → MoveIt collision-free path planning → physical execution; Digital Twin (MoveIt + Gazebo) with real-time bidirectional sync.
- Kinesthetic learning (teach-by-demonstration waypoint recording/replay), predictive maintenance via servo temperature/voltage/current monitoring, dual-camera 3D spatial mapping, and AWS IoT Core cloud migration for multi-robot fleet telemetry.

AgriSmart — AI-Powered Farm Management Platform

[Solo Developer](#)

Tech: React 19, TypeScript, Gemini 2.5 Flash, RAG, ChromaDB/Pinecone, Google Maps API, OpenWeatherMap

- RAG-powered conversational AI grounding Gemini 2.5 Flash in real-time farm telemetry and 1,000+ localized data points; multimodal diagnostics fusing voice, images, soil, and weather data.
- Voice-first UX in 6 regional languages for low-literacy rural users; client-side image compression and ESM-modular architecture optimized for rural edge networks.

AIVA — RAG-Based AI Voice Assistant

Solo Developer

Tech: Python, Ollama (DeepSeek Coder), ChromaDB, MongoDB, Flask, Docker Compose, Tesseract OCR, Google Speech API

- RAG system with ChromaDB + DeepSeek Coder v2: 10+ file format ingestion, OCR fallback (Tesseract/OpenCV), code-aware recursive chunking, source citations; containerized via Docker Compose managing multi-service orchestration.
- Voice-first interface with automated script execution engine (Python/PowerShell/Bash) and built-in security sandboxing; full-stack Flask API with local LLM and vector DB orchestration.

Autonomous Drone Navigation — Deep Reinforcement Learning

Solo Developer

Tech: Python, PyTorch, Microsoft AirSim, OpenCV, TensorBoard, CUDA

- Double DQN agent with N-step returns + Hindsight Experience Replay; attention-based sensor fusion (9 LiDAR + 5 cameras → 120-dim state vector) for 360° awareness; stable convergence over 10,000+ training episodes.
- Curriculum learning across 11 flight skills with adaptive difficulty and curiosity-driven exploration (prediction error models) for significantly improved sample efficiency in sparse-reward environments.

AI Data Scientist Copilot

Solo Developer

Tech: Python, Streamlit, Google Gemini API, Pandas, SciPy, Matplotlib

- NL-to-code analytics platform mapping user queries to statistical methods with 90%+ accuracy; 7+ modules (OLS regression, IQR outlier detection) auto-generating context-aware visualizations; handles CSV, Excel, JSON with intelligent type inference and missing value imputation.

GATE Nexus — AI-Powered Exam Preparation Platform

Solo Developer

Tech: React 18, Node.js, Express, MySQL (8-table schema), Google Gemini API, PowerShell

- Full-stack platform with embedded Gemini AI tutor, YouTube video-completion tracking via chunked bulk SQL inserts, and real-time analytics dashboard across 16 years of PYQs and 50+ syllabus topics with sub-second load via optimized JOINS.

ACHIEVEMENTS, LICENSES & CERTIFICATIONS

Key Achievement

- GATE DA (Data Science & Artificial Intelligence) — Qualified | National-level competitive examination

Certifications (21 total — selected highlights)

1. Machine Learning with Python: Linear Models to Deep Learning — MIT (Pursuing)
2. CS50's Introduction to Programming with Python — Harvard University
3. GDSC ML Bootcamp — GDSC MMCOE | Machine Learning, Pandas
4. AI Builders Lab Bootcamp
5. Agile Scrum in Practice
6. Python with Advanced AI — GUVI / Skill India Digital (IIT Madras validated)
7. Deep Learning for Developers — Infosys Springboard
8. Generative Models for Developers — Infosys Springboard
9. OpenAI Generative Pre-trained Transformers — Infosys Springboard
10. Computer Vision 101 | Introduction to Deep Learning | AI Primer — Infosys Springboard